



East West University

CSE 475 (Machine Learning)

LAB-1

Machine Learning Workflow & Data Preparation

Course instructor

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Objective: To understand and apply data preprocessing techniques by identifying and handling missing values, inconsistent data, duplicate records, and noisy data to produce a clean dataset suitable for analysis.

Sample Dataset:

Customer ID	Age	Monthly Income	Credit Score	Loan Amount	Previous Loans	Loan Approved
1001	25	35000	720	100000	1	Yes
1002	32	45000	680	150000	2	No
1003	NULL	40000	700	120000	1	Yes
1004	28	38000		110000	0	No
1005	35	55000	750	200000	3	Yes
1006	40	NULL	690	130000	2	No
1007	29	42000	710	125000	1	YES
1008	31	47000	730	140000	2	No
1002	32	45000	680	150000	2	No
1010	-15	50000	740	180000	2	Yes

Duplicate Data (points to row 1002 and 1002)

Inconsistent Data (points to -15)

Null Data (points to NULL)

Missing Data (points to empty cell)

Missing Data: Missing data refers to values that are **not recorded or not available** in a dataset. It occurs when no value is entered for a variable in a record. It is often represented as **blank cells, NULL, or N/A**.

Null Data: Null data is a special type of missing data where the value is explicitly marked as **NULL**, meaning the data is unknown or does not exist. It indicates absence of a value in the database.

Inconsistent Data: Inconsistent data refers to data that is **incorrect, invalid, or not in a standard format**, even though a value is present. It often occurs due to human error or different data entry formats (e.g., YES/yes/Yes, negative age like -40).

Duplicate Data: Duplicate data occurs when the **same record appears more than once** in a dataset. It means identical or repeated entries exist, which can distort analysis results.

Tasks:

- 1. Using prompt, Create Your own dataset with 100 samples containing:**
 - Missing values
 - Inconsistent data
 - Duplicate records
 - Noisy/outlier data
 - A binary class label (Yes/No or 0/1)
- 2. Identify all data quality issues in the dataset.**
- 3. Apply suitable data cleaning techniques:**
 - Handle missing values
 - Fix inconsistent data
 - Remove duplicates
 - Treat noisy/outlier values
- 4. Generate a final clean and noise-free dataset.**

Submit a Report (Doc/PDF) which should contain

- **Details of the assignment**
- **Google Colab Link**
- **Uncleaned and Cleaned version of the dataset**